

EAL/2: Executable engineering arguments and claim-scoped delivery to language models

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Abstract

Engineering conclusions depend on observations, the question asked of a method, its operating conditions and challenges to the inference. EAL/2 represents these dependencies in a versioned argument language with typed methods and a bounded support–attack evaluator. We describe that profile, its claim-scoped host and finite reviewed-family route, and distinguish implementation checks from recipient trials. In a paired 180-call diagnostic on five exposed synthetic tasks, strict raw-only correctness was 4/10 for EAL and 5/10 for meaning-equivalent JSON. Two earlier 48-call pilots gave 10/12 exact recipient answers with a checked packet in each run, although the host had supplied the status. In two new developmental cross-model cohorts, 315 scheduled cases generated 365 model calls. Raw and instruction-only exact statuses ranged from 1/9 to 12/12 by model and cohort. The EAL-host and independent equal-checker arms each retained the oracle-matched final status by construction, but recipients sometimes contradicted it or broke the output contract; a corrected route yielded 51/63 consistent statuses and 11/63 agent-judged faithful explanations. In a separate synthetic Kubernetes revision study, seven author-predeclared operating statuses matched execution and lexical candidate precision at three was 0.5714 on ten author-labelled queries. A 36-call synthetic family-selection study yielded two task-level false supports from authorised but irrelevant choices. The data support bounded inspection and delivery of specified calculations. They establish neither an EAL-specific advantage nor independently validated explanation fidelity, broad task coverage or physical truth. Total cost per correctly accepted, faithfully explained decision remains unmeasured.

Keywords: engineering argumentation, domain-specific language, defeasible reasoning, language models, empirical software engineering, evidence

1. Introduction

An engineering decision can turn on a detail that a plausible summary omits. A vibration record may show a value below a limit, yet its calculation can use a different origin from the claim’s formal question. A challenge to an argument can depend on the very claim it challenges, leaving the dispute unresolved until independent grounds are supplied. A status label without its dependencies is difficult to review or revise.

EAL/2 represents claims, observations, assumptions, reasoning methods and objections as named, scoped declarations. Its evaluator checks supplied evidence and method outputs, then computes the status of the resulting support–attack structure. A text model may author or interpret that structure, but its fluent account does not perform the evaluator’s checks. The engineering question is therefore twofold: what can the language and evaluator check, and which parts of their result survive transfer to a model recipient?

This article contributes a precise account of EAL/2’s implemented profile, a worked case that distinguishes numerical success from a matching formal query, and an analysis of retained model experiments. The relay comparison tests hand-offs from an already correct tool-enabled producer. The notation diagnostic introduces wrong proposed answers and a paired EAL–JSON comparison. Later recipient pilots supplied a checked status to one model on selected synthetic states. The newer host and reviewed-family implementation changes the delivery boundary and motivates the separate cross-model comparison reported below. None of these trials measures model construction of a new engineering argument; neither an advantage caused by EAL syntax nor reliable model preservation of qualifications has been shown.

2. Language and executable meaning

2.1. Design boundary

EAL/2 is the repository’s sole supported source language. Its grammar requires the header `language "EAL/2";`; earlier EAL measurements retain their historical versions and cannot be relabelled as EAL/2 results. The ANTLR grammar recognises source, which is lowered and checked against a typed internal representation before evaluation. A `reasoning` declaration selects one versioned method, for example `engineering/rms/1`. The tool’s

`mode` instead describes collection variability. Built-in and installed methods use the same contract and result-binding checks. Reusable patterns have typed parameters, closed lexical scope and a finite expansion into ordinary arguments; expansion retains the identity of shared observations [1].

The language makes an authored inference reviewable, rather than turning prose into a proof. A claim has a statement and an environment. Evidence declares its acquisition tool, kind, scope, age limit and value predicates. Assumptions add validation evidence and optional time bounds. An argument names a conclusion, a reasoning step and the evidence, assumptions or premise claims on which that step depends. An objection names the exact claim, argument, method, assumption or other objection it challenges. Toulmin's separation of grounds, inference and rebuttal informs this vocabulary [2]. SACM defines a metamodel and graphical notation for structured assurance cases [3]. EAL/2 specifies executable method contracts and status evaluation for its own bounded engineering-argument profile; neither specification's existence establishes that one is easier to use. EAL/2 does not implement the complete argument-construction, contrariness or preference machinery of ASPIC+ [4].

2.2. A query that a numerical threshold cannot settle

Listing 1 is a complete, existing EAL/2 programme from the repository. It asks whether recorded root-mean-square (RMS) pressure about zero is at most 5 kPa for a named sample episode. The external method is identified by a versioned contract; the claim's proposition fixes the subject, quantity, unit, scope, time and query. The `binding` clause requires the observation used by the method to answer that formal query.

```
language "EAL/2";
environment lab { require "site" == "bench"; }
tool readings { version "1"; mode deterministic; }
evidence series {
  tool readings;
  kind measurement_series;
  environment lab;
  max_age 3600;
  require "schema" == "EAL/typed-input/1";
}
reasoning rms_method {
  method "engineering/rms/1";
  rationale "Compute RMS deviation from zero over the supplied pressure
    samples.";
```

```

}
claim bounded_rms {
  statement "RMS pressure relative to zero is at most 5 kPa for this
    sample.";
  environment lab;
  proposition {
    subject "pump-A";
    quantity "pressure";
    unit "kPa";
    scope "sample-episode-v1";
    valid_from "2026-09-23T10:00:00Z";
    valid_until "2026-09-23T11:00:00Z";
    query {"origin":0};
    result "rms" <= 5;
  }
}
}
argument sampled_rms {
  conclusion bounded_rms;
  reasoning rms_method;
  evidence series;
  binding series;
}

```

Listing 1: Complete EAL/2 RMS example; reproduced from examples/rms.eal at the repository revision stated in the data availability section.

The diagnostic’s related vibration task supplies samples 0.3 and 0.4 m/s. With origin zero, the computed RMS is $\sqrt{(0.3^2 + 0.4^2)}/2 \simeq 0.35355$ m/s, below the 0.36 m/s claim limit. A counterexample retains those samples but records an origin of 0.1 m/s. It yields $\sqrt{((0.3 - 0.1)^2 + (0.4 - 0.1)^2)}/2 \simeq 0.25495$ m/s. That smaller result answers a different query. The checked claim is **unsupported** for that supplied observation, even though either scalar passes the numerical threshold [5]. The interpreter’s result is conditional on the authored question, observation identity and method implementation; it does not certify that the physical measurement or model is adequate.

2.3. Support, objections and bounded evaluation

Let $E(a)$, $A(a)$ and $P(a)$ be an argument’s evidence, assumptions and premise claims. Local usability checks declared context and time, evidence acquisition identity and freshness, required predicates, and the method’s typed input and output. The authored argument asserts a conditional relation between those checks and its conclusion. It may have alternative derivations; an observation reused by several arguments retains one identity. The

evaluator rejects cyclic claim-premise dependencies and bounds programme size and graph relationships [6].

Objections form a support–attack graph, including objections against objections. Write $U(n)$ for a node’s local usability, $P(n)$ for its required claims, $A(n)$ for attackers, and $D(c)$ for arguments deriving claim c . Starting with undecided labels, the evaluator monotonically adds justified accepted or rejected labels to its least-information fixed point:

$$\text{Acc}(n) \Leftarrow U(n) \wedge \bigwedge_{c \in P(n)} \text{Acc}(c) \wedge \bigwedge_{a \in A(n)} \text{Rej}(a), \quad (1)$$

$$\text{Rej}(n) \Leftarrow \neg U(n) \vee \bigvee_{c \in P(n)} \text{Rej}(c) \vee \bigvee_{a \in A(n)} \text{Acc}(a), \quad (2)$$

$$\text{Acc}(c) \Leftarrow \bigvee_{n \in D(c)} \text{Acc}(n), \quad \text{Rej}(c) \Leftarrow \bigwedge_{n \in D(c)} \text{Rej}(n). \quad (3)$$

The claim-premise graph is acyclic, while attacks and objection-support relationships may cycle. This is a defined conjunction/disjunction extension whose attack-only restriction agrees with grounded labelling for a finite Dung graph [7, 6]. It does not generate objections from natural-language contradiction or rank competing methods.

For a usable argument, acceptance gives **supported**; a rejected or undecided derivation with no uncontested alternative gives **contested**. With no usable derivation the claim is **unsupported**; a failed environment check gives **out_of_scope**. These labels describe the supplied argument calculus. In particular, **unsupported** does not prove a proposition false. Method execution failure, an unmatched query and an unavailable observation remain distinguishable in the assessment record. Unit, time and scope checks reduce accidental reuse of a computation; engineers still choose the relevant quantities, assumptions and observations.

2.4. Claim-scoped host and reuse boundary

The later implementation moves an already authored argument into a host-owned catalogue. An operator pins source and method digests, context, assessment time and permitted claims. A trusted launcher authenticates the recipient’s principal, validates finite exact claim identifiers, snapshots its grant sets and creates a separate MCP server exposing only assessment, explanation and completion for granted artefact–claim pairs. Recipient prose cannot introduce an ungranted source or claim; a pinned host invocation fixes

the exact pair. A text-only model can request a schema-validated operation through the application host; the host, rather than the model, initiates MCP and executes the registered tools. The result of `finish` is recomputed against the saved assessment and remains the application status even if recipient prose contradicts it [8].

The bounded claim packet (schema `eal2-claim-packet/2`) returns one status, source and method identities, context fingerprint, assessment and collection identifiers, and a short account of the selected argument, failed dependency or active objection. It limits the displayed arguments and objections and marks truncation; an authorised `explain` call retrieves a larger claim-specific direct trace. That trace identifies premise claims but does not expand all transitive premise evidence. A packet is at most 3,072 UTF-8 bytes and the explanation at most 65,536. Those limits constrain transport, not the number of semantic checks performed: the current `assess_claim` path collects for the whole artefact before selecting one claim. An assessment ID alone is not an access grant. The recipient MCP route stores a principal-bound issuance record that survives server restart; the in-process task-family route instead retains issued IDs per instance and requires reassessment after its restart [8, 9]. A saved `finish` recovers a historically checked status; a claim about the current cluster needs fresh collection and reassessment. Source and observation digests detect a changed represented record; the packet explicitly labels evidence integrity `consistency_checked_not_authenticated`. Physical sensor identity, collector integrity and truth of an authored warrant require separate evidence.

A host-owned family catalogue groups finite, typed parameter tuples, each mapped to an exact pinned artefact and selected claims. Its loader checks that each parameter has a distinct context path and a corresponding equality constraint in the selected claim’s EAL environment. A recorded reviewer label and contract digest bind the declared review to the source, method, context, time and claims; the software cannot authenticate the reviewer or show that the review was independent. An unlisted tuple cannot inherit an existing verdict. Candidate retrieval ranks authorised families by lexical overlap with task text. It returns names and matched terms, without choosing parameters, granting access or asserting applicability. A trusted application then chooses explicit bindings and a claim, and `TaskFamilyHost` checks the family, artefact and claim grants before assessment [10, 9]. This avoids treating semantic similarity as an inference rule, at the cost of reviewing a new tuple when scope changes.

The architecture separates three possible errors. A source can misstate the original engineering question; an observation can be stale, counterfeit or irrelevant to that source; and a recipient can misreport an otherwise checked result. Host execution addresses specified checks in the second and third stages. It cannot infer omitted resource contention, model inadequacy or missing defeaters. Even an authorised, exact family can address a different question from a vague task: a supported status for a wrongly selected family is false support relative to that task, without a grant bypass. Applicability of a selected family to the user’s intent needs trusted review. A deployed service additionally needs an authenticated endpoint that binds the principal and grant; the present trusted launcher and in-process route define that boundary but do not demonstrate arbitrary remote identity integration. Family candidate retrieval and exact binding are application APIs and are not yet exposed through the recipient MCP endpoint. Similarly, existing targeted objections and finite fixed-point labels do not amount to ASPIC+ construction, contrariness and rule preferences. A demand for implicit attacks or competing default priorities would warrant a separately named, versioned reasoning-method implementation and tests before any such claim of conformance [4, 6].

2.5. Resource contention as an inferential change

An additional synthetic EAL/2 example represents a checkout-api deployment in the payments namespace on a shared Kubernetes node. It links a declared image digest and attestation record to a reported rollout, a five-minute `checkout-create` trial at 200 requests per second, 10,000 recorded outcomes, four sampled p99 latency values and a snapshot of requested CPU headroom. A Wilson lower bound and a finite temporal predicate check specified numerical conditions. The bounded canary claim concerns the recorded trial. A separate operating-basis claim proposes considering the same build for the next fifteen minutes under unchanged traffic and resource conditions; that future performance is an authored, defeasible inference, not a consequence proved by the sampled result [11].

The first engineer’s evidence must connect source commit, build attestation, image digest, rollout generation, ready endpoint identities and observed requests and responses to the `*same*` service episode. An assessor then needs coverage and units for the latency series, a defended sampling model for the Wilson calculation, and a record of request accounting, actual usage and

Table 1: Claim delivery stages and the decisions each stage is permitted to make. Lexical nomination cannot authorise execution; recipient prose cannot replace a host status.

Stage	Supplied material	Authority and output
Nominate	Task text, permitted family metadata	Lexical suggestions; no binding or verdict
Resolve	Explicit family, typed tuple, claim and caller grant	Trusted application checks exact reviewed case and grants; pins artefact
Assess	Pinned source, context, method, observations and time	Host checks represented evidence and emits a claim status, IDs and short reason
Explain and finish	Principal-bound assessment ID and claim	Authorised direct trace; saved host status remains final
Communicate	Checked packet and task text	Recipient prose is retained separately and scored for fidelity

throttling. Kubernetes CPU requests affect placement, whereas CPU limits are enforced through throttling; a request-headroom snapshot does not bound a co-tenant’s actual consumption [12]. After a second engineer deploys a batch job, a record of the batch Pod and target sharing a node, with time-aligned CPU usage, target throttling, queueing and latency, supplies a specific challenge to the operating inference. A controlled batch-admission or isolation trial under matched traffic would strengthen causal attribution. A current cluster query alone cannot prove future p99 or exclude unobserved interference.

For the example’s three authored revisions, the operating-basis status changes from **supported** to **contested** when the new batch observation activates an objection against the reasoning step, then to **supported** with a fresh trial and complete monitor nonfinding in a later record. That sequence alone does not show that the stated mitigation caused the change. The historical canary remains a bounded report of its earlier sample; its episode is identified by retained assessment inputs and time, not the recurring claim ID alone. Dropping the contention record makes the raw evaluator support the operating basis because no attack is submitted. A fixture-specific application

gate retains that raw label but returns unresolved. Generic `assess_claim` does not invoke this gate, so a production caller must attach an equivalent mandatory-acquisition policy before treating the raw status as current assurance. Before acceptance it checks the mandatory records’ source, request, time and matching manifest trial, Pod and node identifiers; missing, stale, failed or incoherent monitor records fail closed. The response and p99 record envelopes now carry trial, Pod/node, window and series identifiers plus a payload digest; the gate recomputes and matches these against the manifests. Those producer-declared bindings check record coherence but do not authenticate the collector, prove physical-series origin or establish that the query observed all interference. An operational deployment needs separately verified acquisition coverage and event-triggered reassessment when the work mix or resources change [11, 13]. This counterexample concerns evidence completeness and the source-to-world relation. A more elaborate argumentation solver cannot infer an omitted event from silence.

3. Evaluation design

3.1. Questions and materials

The design claim is that EAL/2 can express and execute the specified engineering dependencies. Executable examples and regression checks exercise that claim within the implemented profile; their pass count does not establish broad representational completeness or human readability. The model questions are different. Does EAL syntax, compared with meaning-equivalent JSON, improve a pinned recipient’s status answers? Can relevant raw observations correct a wrong proposal without damaging a correct qualified proposal? Does supplying an interpreter conclusion change the answer? A rival explanation of any syntax difference is a prompt-surface effect involving length, tokenisation, familiarity or output compliance, rather than a general reasoning gain [14].

We distinguish the earlier relay pilot from the later notation diagnostic. The relay programme used six previously exposed synthetic tasks, two repetitions, 27 model/transfer conditions and 324 attempted endpoints; 323 completed. In its declared primary contrast, the same GPT-4.1 tool-enabled producer supplied already correct answers on twelve blocks to a GPT-4.1 nano recipient. With tool evidence the recipient retained 12/12; with the producer’s answer alone it retained 8/12. The descriptive task-weighted difference was 33.3 percentage points, with a task-cluster interval of 0.0–66.7

points and an inconclusive prespecified decision. There were no producer errors in those twelve blocks, so this contrast has no correction denominator [15].

The later experiment used five exposed synthetic tasks: a circular defence, an independently grounded defence, a registered RMS calculation, a wrong-origin RMS observation and a sampled negative finding. A source-repair task was excluded because its interface could not be meaningfully matched in this diagnostic. Each task had eighteen conditions and two repetitions, giving 180 scheduled and attempted model calls. The recipient was snapshot `gpt-4.1-nano-2025-04-14`, at temperature zero with a 2,048-token output limit. Each call used a fresh message context, no tools and no repair. EAL and JSON packets had equal parsed meaning, observations and common reference; their natural input lengths differed and the reference used EAL vocabulary. A correct or deliberately incorrect proposed answer was fixed for each task/repetition stratum. The crossed views were answer only, relevant raw observations, irrelevant observations for another assembly, and raw observations plus interpreter conclusions. A fifth view supplied raw observations without a proposed answer [16, 17].

The experimental unit for a notation comparison is a matched task/repetition pair. Two repetitions of the same exposed template are dependent and sometimes contain different wrong proposal labels; they are not ten independent task types. There was no protected unseen-task sample. Assignment, model settings, prompts and references were frozen before generation, with a pre-generation amendment to completion-status scoring and the available-information reference. After a known-cost output-limit failure at call 30, an operational continuation finished the original schedule without changing prompts or retrying that failure. Its revised stop handling was documented after the event [18]. This departure and task exposure prevent confirmatory population inference.

3.2. Outcomes and estimands

The strict primary score admits a response only if its whole text is a JSON object with exactly `claims` and `basis`, all requested claim identifiers and allowed statuses, and a list-valued basis. Full-information correctness $Y_{ik}^F(n, v, a)$ requires all statuses to match the original complete-observation reference for task i , repetition k , notation n , view v and proposal a . Available-information correctness Y^A compares the same response with the frozen interpreter reference for the records actually supplied. These references co-

incide for raw-only, relevant-raw and assessed views. For answer-only and irrelevant-record views the interpreter has no relevant observation, so asserting support can agree with the hidden full answer while being unjustified by supplied information.

The finite paired notation contrast is

$$\hat{\Delta}_N = \frac{1}{5} \sum_{i=1}^5 \frac{1}{2} \sum_{k=1}^2 \left[Y_{ik}^F(\text{EAL}, \text{raw}, \emptyset) - Y_{ik}^F(\text{JSON}, \text{raw}, \emptyset) \right]. \quad (4)$$

The corresponding raw-evidence contrast holds the wrong proposal, task and notation fixed and compares answer + raw with answer. It is a response difference under this prompt, not a naturally occurring error-correction rate. Every failed or malformed attempt remains in the correctness denominator. A detected false-support event means that an admitted response marks at least one requested claim **supported** when the selected reference does not. Malformed text and provider errors are excluded from that event count, so its zero would not establish absence of false assertions in unadmitted text.

For an admitted claim map m and a reference o , define $\text{Matches}(m, o) = (m \neq \emptyset) \wedge \bigwedge_{c \in \text{Ids}(o)} (m[c] = o[c])$. Define $\text{FalseSupport}(m, o) = (m \neq \emptyset) \wedge \bigvee_{c \in \text{Ids}(o)} (m[c] = \text{supported} \wedge o[c] \neq \text{supported})$. Algorithm 1 summarises the retained strict scorer and matched contrast procedure using these two exact tests. The implementation additionally verifies frozen schedule and record integrity. The algorithm distinguishes the two reference questions and never repairs malformed model text.

Algorithm 1 Strictly score retained trials against both references and pair notation outcomes

Require: Frozen task/repetition schedule S , attempted records R , full references F , view-specific references V

Ensure: Correctness indicators Y^F, Y^A , detected false support, complete-pair contrasts

```

1: for all scheduled row  $s \in S$  do
2:    $r \leftarrow \text{FINDATTEMPT}(R, s)$ 
3:    $C_s \leftarrow \text{CLAIMIDS}(F_s)$ 
4:    $m \leftarrow \text{ADMITSTRICTCLAIMMAP}(r, C_s)$ 
5:   for all reference  $o \in \{F_s, V_s\}$  do
6:      $Y_s^o \leftarrow \text{MATCHES}(m, o)$ 
7:      $H_s^o \leftarrow \text{FALSESUPPORT}(m, o)$ 
8:   end for
9: end for
10: for all matched task/repetition block  $b$  do
11:   if both raw-only notation attempts are present then
12:      $d_b \leftarrow Y_{b,\text{EAL},\text{raw}}^F - Y_{b,\text{JSON},\text{raw}}^F$ 
13:   end if
14: end for
15: return  $Y^F, Y^A, H, \{d_b\}$ 

```

All 180 calls were attempted once. Eighteen responses failed the exact schema and one hit the output limit; every response retained model identity and usage. The summed reported usage was 1,919,842 input tokens (1,837,824 cached) and 29,816 output tokens. Configured-rate cost was USD 0.0660738, an estimate rather than an invoice. The archive stores raw records, hashes, scoring tables and the continuation record [17].

4. Results

Table 2 retains the attempt denominator and separates the full and supplied-information references. The strict raw-only notation comparison yielded EAL 4/10 and JSON 5/10, or $\hat{\Delta}_N = -0.10$. Four matched pairs were correct in both formats, one only in JSON and five in neither. No task showed an EAL-only success; the single discordant pair arose in the sampled-negative-finding task. This describes these ten paired attempts and does not estimate a population JSON advantage.

Table 2: Strict complete correctness, with ten attempted calls per notation and view. Correct and incorrect describe the injected proposal against the full-information answer.

View and proposal	EAL full	JSON full	EAL available	JSON available
Raw only; no proposal	4/10	5/10	4/10	5/10
Correct; answer only	10/10	9/10	2/10	2/10
Correct; relevant raw	2/10	7/10	2/10	7/10
Incorrect; answer only	0/10	2/10	4/10	4/10
Incorrect; relevant raw	2/10	5/10	2/10	5/10
Incorrect; assessed	7/10	8/10	7/10	8/10
Correct; irrelevant	6/10	6/10	0/10	0/10
Incorrect; irrelevant	2/10	3/10	3/10	0/10

With a deliberately wrong proposal, relevant raw records produced two EAL and five JSON complete corrections, compared with zero and two for answer-only. Supplying interpreter conclusions along with the raw records raised complete correctness to seven and eight. These differences involve supplied information as well as changes in prompt length, and cannot isolate independent calculation by the recipient. In the correct-proposal stratum, EAL answer-plus-raw scored only 2/10 versus 10/10 for answer only; JSON scored 7/10 versus 9/10. Under the supplied-information reference, answer-only correctness was 2/10 for a full-information-correct proposal in either notation, because the proposal itself supplied no observation. Describing that reference change as damage to correct empirical support would invert the evidence requirement.

The irrelevant-record control exposes a separate failure. Across its four representation/proposal arms, only 3/40 attempts were correct for information actually available, and 26/40 contained detected false support in admitted responses. None of the 28 proposals wrong under that reference was corrected. Values from another assembly could appear numerically plausible, but they were not observations for the requested scope. In raw-only trials both representations failed the circular-defence and wrong-origin tasks in both repetitions; each succeeded on the independently grounded defence and matching RMS task twice [17].

The prespecified strict output contract counts the eighteen malformed responses as failures. A later, explicitly exploratory readout accepts a strictly parsed raw JSON claim map with extra top-level fields and ignores the `basis` shape. It changes raw-only results to EAL 4/10 and JSON 6/10. It

also increases detected false-support events across all conditions from 69/180 to 77/180. Fence-only normalisation changed no primary count. Relaxing output shape therefore cannot reveal an EAL advantage in this sample; it changes the measured endpoint and was chosen after schema failures had been seen [17].

5. Host delivery and its present evidence

5.1. Earlier checked-packet recipient pilots

Two subsequent developmental runs each made 48 one-shot requests to the same GPT-4.1 nano snapshot. Four selected synthetic roots had three related evidence states each. Every state was rendered as raw EAL, an EAL-derived JSON graph, EAL with an application instruction, or an earlier checked host packet. This packet predated the claim-scoped /2 packet and the principal-bound family route in Section 2.4; neither new feature was exercised by those model calls. The independent brief-level reference and interpreter agreed for the twelve states, but no second masked source review or independently implemented equal checker was part of the live blocks [19, 20].

Table 3: Exact recipient status and format on twelve related states in each earlier 48-call developmental run. Counts are descriptive; the checked packet already supplied the host’s answer.

Route	First exact	First false	Repeat exact	Repeat false
Raw EAL	3/12	2/4	6/12	2/4
Derived JSON	8/12	3/4	8/12	3/4
Application instruction	5/12	1/4	5/12	1/4
Checked packet	10/12	0/4	10/12	0/4

The earlier checked packet required 3,754 recipient input tokens over twelve calls, compared with 17,855 for raw EAL in each schedule. Those are reported provider tokens, not total workflow resources. The recipient still contradicted the host on two supported inspection states in each run; a correct application result therefore retains the host’s status rather than adopting the recipient’s field. The first packet also produced at least one correct label with an inaccurate causal explanation under subsequent qualitative inspection. No blind explanation score, acquisition cost, host latency

or authoring effort was recorded. The JSON graph was generated from EAL source rather than independently authored, and some of its input tokens were cached. These runs estimate delivery on four selected roots, not the gain from the new host implementation or from EAL notation itself [19, 20].

5.2. New implementation checks and cross-model comparison

Two developmental schedules crossed five routes with three configured model classes: raw source and records; the same input with a short skill instruction; a model-originated JSON assessment request through the host followed by a possible second call; the compact claim-scoped EAL packet; and the same final-answer contract using an independently implemented generic graph checker. The strong reasoning-model route is the raw condition on that model class. The request route validates a model-selected artefact and claim against a fixed allowed set; the correct candidate description was placed first in every route brief, so this tests menu/request syntax rather than family retrieval or new parameter bindings. The previously exposed four-root corpus supplied 180 first-call scenarios and a separate, singly authored three-root cohort supplied 135. All source, observation, prompt, checker and order identities were frozen before calls. An initial 180-case freeze stopped after three attempts (two completed and one returned-model identity mismatch); those attempts and their USD 0.0050428 configured-rate cost remain a separate diagnostic. Revised freezes allowed the observed dated nano identity and completed 180/180 and 135/135 scheduled cases without a retry or failed provider attempt [21, 22].

Offline, the generic checker and EAL reference agreed on 12/12 scored states from the older pilot and 9/9 from the new cohort. Five newly authored challenge pairs passed in both systems. Sixty generated older-pilot and 45 new-cohort envelope mutations exercised the checker alone, so those mutations do not establish EAL/checker equivalence. The new cohort has no independent human reference, source or comparator review. The resulting parity establishes agreement on selected represented cases; it neither certifies the reference against real engineering systems nor measures model reasoning. For both checked routes, the final host status inherits the oracle-matched reference by construction. Thus every checked final status matches that reference and none finally falsely supports an adverse state, but this is an enforcement property of the runner. Recipient format and explicit agreement with that status are distinct outcomes [22].

Table 4: Developmental cross-model calls. Raw and skill columns count strict exact recipient statuses. EAL-host and equal columns count strict recipient JSON with a status consistent with the host-owned answer, before explanation review; their final application status is oracle-matched by construction. Request counts are valid first-turn host requests, not valid final answers.

Cohort	Model	Raw	Skill	EAL-host	Equal	Request
Exposed, 12 states	Nano	3/12	2/12	8/12	6/12	3/12
	Luna	10/12	10/12	8/12	11/12	12/12
	Sol	10/12	12/12	12/12	12/12	12/12
New, 9 states	Nano	1/9	1/9	0/9	3/9	5/9
	Luna	8/9	8/9	5/9	9/9	9/9
	Sol	7/9	7/9	9/9	9/9	9/9

The exposed cohort had two false-support outputs in raw and two with the skill (one each for nano and Luna in each arm); the new cohort had none in those arms on its selected adverse states. For Luna, the EAL packet elicited a status contradicting the host in four of twelve valid pilot responses and four of nine valid new-cohort responses, whereas the equal-checker packet had one and zero contradictions. Nano returned malformed EAL responses in 4/12 and 8/9 cases, and malformed equal-checker responses in 6/12 and 6/9 cases. Sol gave valid, consistent checked responses throughout. These differences describe selected related states and unequal packet contents, not evidence of a notation-specific mechanism. They also show why the host-owned final status cannot be replaced by the recipient’s prose [22].

The request arm elicited 27/36 valid first-turn host requests in the exposed cohort and 23/27 in the new cohort. Its frozen second turn retained a system instruction demanding another operation request while the new user turn demanded final claims and explanation. All 63 final answers failed the strict final schema, including those after successful host calls; this is a prompt-conflict diagnostic, not a clean test of routed delivery. A corrected route was separately frozen and run. The 180-case run made 207 model calls: 257,339 input and 25,359 output tokens. The 135-case run made 158 calls: 177,085 input and 17,407 output tokens. Summed per-scenario API times were 1,923.205 and 1,448.013 s. Median per-scenario latency by model ranged from 7.68 to 11.82 s and from 7.92 to 11.59 s, respectively, including routed second turns. At configured rates, model charges were USD 0.3798973 and USD 0.2558221, or USD 0.6357194 combined. These are provider usage

and call timings, not invoices, wall time, cluster-acquisition cost or all-in cost [22].

Two later, separate route-only freezes corrected the second-turn instruction while retaining the same fixed first-turn menu and synthetic cases. Across 36 exposed-root and 27 new-root attempts, 51/63 requests produced an accepted host assessment and a strict recipient answer whose status agreed with the host: nano 9/21, Luna 21/21 and Sol 21/21. The twelve other nano first turns were rejected as nine incorrect or unauthorised requests and three malformed requests, not safe negative decisions. All 51 accepted statuses matched the singly authored reference, including seventeen adverse states, because the host supplied the checked answer; these are recipient-consistency counts before explanation review. The two revised runs made 114 model calls, with 83,911 input and 5,866 output tokens, zero provider failures or retries, and USD 0.0976742 in configured-rate model charges. Median per-case API times by nano, Luna and Sol were 11.179, 18.507 and 18.589 s. The first-turn samples are separate draws, so the change from the original 63 malformed final answers diagnoses the prompt procedure rather than a learned recipient improvement. It still tests a prefixed fixed menu, not candidate retrieval, typed family selection or native MCP tool use [23].

An exploratory post-call review sampled one state from each of seven roots across fifteen model-route conditions (105 outputs from the original flawed route schedule and four other arms). Two separate de-identified AI review passes used the same single-author dossiers. The adjudicated categories were 49/105 faithful, 13 incomplete, four materially false and 39 malformed or absent; reviewers agreed on the faithful binary for all 105 and differed on one non-faithful category. Judged faithful counts by route were raw 16/21, skill 15/21, EAL-host 4/21, equal checker 14/21 and the flawed request route 0/21. The subset rule was documented after some outputs were observed, and response text could reveal an arm despite withheld linkage. The reviewers saw a full task dossier rather than each recipient’s exact exposed input. A cautious statement about missing input could therefore be penalised despite reflecting the recipient’s actual view. The compact packet also omits some transitive premise evidence. These are exploratory end-to-end explanation-completeness signals on dependent synthetic states, not human adjudication, a model-fault attribution, or a complete 315-case fidelity rate. They cannot supply the denominator for all-in cost per correct accepted decision [24].

A separate exploratory review assessed all 63 corrected route finals against a rubric fixed before those outputs were scored. Its packet retained the

actually delivered host packet alongside author source and observations, while withholding model linkage. Two independent AI reviewers agreed on 59/63 exact categories and 61/63 faithful-binary labels; a third agent resolved four differences to 11 faithful, 38 incomplete but nonmisleading, two materially false and twelve malformed or absent. Thus all 51 accepted recipient statuses matched the host, yet only 11/63 route attempts (11/51 accepted answers) met this strict evidence-level explanation rubric. Generic claims that the host accepted a route often omitted a decisive record, objection or calculation. The packet itself can omit transitive premise detail, so this is an end-to-end completeness finding on authored synthetic cases, not a causal diagnosis of recipient reasoning or human-validated fidelity. The agent review time omits source authoring, independent verification and live acquisition and cannot establish all-in cost [25].

A separate synthetic Kubernetes host/retrieval study specified seven evidence revisions before its local timing run. It separates a historical test-record claim from a provisional operating-basis claim. The host matched the author’s seven expected operating statuses and decisive causes and seven expected historical-record statuses. A fresh pressure report contested only the operating basis; stale load or monitoring coverage left its earlier record supported but made the operating basis unsupported. A separate reservation report answered the named pressure objection. This corrected case supersedes an earlier fixture in which an attack on a literal report claim conflated a later operational challenge with falsification of the past record. The mean compact packet occupied 1,553 UTF-8 bytes versus 6,042 for its direct trace, a mean ratio of 0.257. In twenty local repeats per revision, median combined assessment, explanation and completion time ranged from 161.0 to 174.4 ms. These timings include local Python and filesystem work, not MCP network transport, live cluster collection or model latency [26].

Ten singly authored candidate queries gave lexical retrieval 8 true positives, 6 false positives and 1 false negative at a limit of three: micro precision 0.5714 and recall 0.8889. Top-one accuracy was 7/8 among queries with a target; one of two no-target queries produced no suggestion. The broad Kubernetes query returned an unwanted candidate, and a traffic-switch synonym missed its intended family. Exact resolution rejected the unlisted `prod_west` tuple and an unauthorised family–claim pair despite candidate suggestions. These are developmental labels from the fixture author, not independent retrieval relevance judgements. Packet size is not token usage, and 7/7 status agreement tests consistency with an authored graph rather than whether the

authored inference or a cluster observation is correct [26].

A distinct developmental selection schedule presented lexical suggestions to three model classes on twelve synthetic tasks, yielding 36/36 completed one-shot calls. The model saw task text, trusted scope and candidates, but no EAL source, evidence revision or checked answer; it selected a family, typed cluster/namespace binding and claim, or abstained. The trusted exact resolver owned the subsequent status. Selection or appropriate abstention was correct for nano on 9/12 tasks, Luna on 10/12 and Sol on 10/12. All three selected the same latency family across baseline, contention and stale-coverage evidence revisions; the host returned **supported**, **contested** and **unsupported**. All three unlisted **prod_west** tuple requests were denied. A missing lexical candidate for a traffic-switch synonym led all three to abstain on a legitimate task. On two ambiguous or broad tasks nano selected an authorised but irrelevant latency family, and its checked **supported** result gave two task-level false supports. This was an applicability failure rather than a grant bypass; Luna and Sol abstained on those tasks [27].

The corrected frozen input digest for this selection study is **b5ef86e16f16d539d75d0632497b73862a6f6ed4c1ea9cd42d6fdb313244c7cc**; an earlier audited freeze was superseded with zero paid calls. The retained archive records 10,938 input and 2,326 output tokens, no retries or failed calls, and USD 0.0195864 in configured-rate model charges. Median per-call API times were 7.339, 8.316 and 8.899 s for nano, Luna and Sol. Its exact freeze and usage ledger were independently audited, but task relevance, source and statuses were labelled by one fixture author. These results demonstrate selected text-only candidate handling and exact denial on synthetic records; they do not establish field selection accuracy, live cluster truth, native MCP model use or faithful explanation [27].

The separate proposed 96-call negative-revision pilot retains its question about brief-to-formal fidelity and exact reversals. Its planned 48-call interpretation stage and 48-call drafting/revision stage still require eight new reviewed roots and a runnable route under the current source and host contracts. The proposed 960-call mechanism study conditions on reviewed source, varies supplied inference stages, and includes 192 deterministic EAL/equal-checker executions; a checked label in a text prompt cannot substitute for a principal-bound host-owned final answer. The proposed 800-call deployment pilot examines authoring, ten consumers per source and cost of reuse across eight roots; the current family catalogue makes exact binding and grant failures measurable but does not itself demonstrate generalisation.

The 800-call design’s proposed candidate negative-finding grammar factor conflicts with the present no-grammar-extension scope. Removing or replacing that factor changes the design and count, so an amended schedule must be preregistered rather than calling the original protocol valid as is. These are distinct specified, unrun investigations. Adapting their schedules to the /2 packet, review contract and independent checker requires a versioned amendment and a new input freeze; historical frozen calls remain historical evidence [28, 29].

In the developmental route comparisons, an adverse event is an admitted **supported** output when the author-specified reference requires another status. A confirmatory endpoint needs an independently reviewed reference. The full decision endpoint also requires the named status, the output contract and a separately adjudicated faithful explanation; malformed answers, failures and retries remain in attempted denominators. The estimated cost per correctly accepted decision would divide model charges, acquisition and host computation, explanation review, and initial authoring and review amortised over the evaluated reuse by the count of faithful, correctly accepted decisions. Its denominator is undefined when no decision qualifies. The reported schedules have neither complete human effort nor recipient-input-matched human explanation adjudication, and provider token prices are estimates rather than invoices. The analysis therefore leaves total cost per correct accepted decision unavailable rather than reporting a token-only proxy [21].

6. Interpretation and limitations

The implemented checks have a clear use: an engineer can ask which observation, query, scope, assumption or challenge caused a status and can reassess after a recorded change. Typed method contracts prevent a numerically passing result for the wrong formal query from silently supporting the claim. That is a software behaviour claim supported by the specified semantics and repository checks. Whether the authored relation from a measurement to a physical conclusion is sound still requires domain evidence. Reusing the same observation through two arguments does not create two independent measurements; the calculus does not assign numerical credibility to agreement among arguments.

What can the model trials establish about the notation itself? In this diagnostic they establish only a finite paired outcome for one model on five exposed tasks. Meaning-equivalent EAL and JSON differ in tokenisation and

length, and the common reference uses EAL terms. Even a larger observed difference in this design would combine those surface mechanisms. Prompt-format sensitivity elsewhere supplies a reason to compare plausible formats; it does not predict EAL’s direction of effect [14]. The present strict and exploratory comparisons give no positive EAL notation result. A fully structured JSON interchange remains a credible design alternative for the same typed representation.

Computed interpreter conclusions are different from raw measurements. Their larger correction counts show that the tested recipient can often use a supplied status; they do not show that it independently applied the method or checked the scope. The earlier relay benefit likewise concerns preservation of a correct producer’s qualified answer. The checked-packet pilots and new cross-model responses show that communication can fail after a correct host assessment. The frozen request-route conflict also demonstrates an application-prompt failure after successful host calls. These observations separate preservation, recovery from injected wrong labels, host execution and recipient communication. A host-owned final result enforces the last boundary for represented statuses, but the engineering warrant and acquisition chain remain independently reviewable.

Several limits affect inference. The five diagnostic task templates and six relay templates were public development material, and related cases share mechanisms. The later packet pilots reused four roots and did not exercise current claim-scoped delivery. The new cross-model cohorts remain developmental: one reuses those exposed roots, while the other has three singly authored roots and lacks masked independent adjudication. Their equal checker has separate code, but selected endpoints were matched offline and supplied to both checked arms; 21/21 final statuses per model on either arm do not estimate recipient reasoning accuracy. Repetitions and evidence revisions are dependent; task-cluster intervals from six clusters have poor coverage in the repository’s sensitivity simulation [15]. A failed schema or output limit counts against an end-to-end answer but does not reveal the model’s latent status judgement. The short public justifications are archived, yet no independently validated quality score of those explanations is available. A correct interpreter answer is conditional on its source, method contracts and reference fixtures, not a measurement of real plant behaviour. The new family resolver checks only finite tuples with a recorded review assertion, and its lexical retriever has ten single-author offline candidate queries and twelve singly authored model-selection tasks, with no independently adjudicated or

field selection accuracy. Finally, the stopped identity diagnostic, request-route prompt conflict, after-call continuation and post-hoc format readout must remain visible rather than being treated as prespecified confirmatory analyses. Repeated adaptive use of an exposed task set can reward local tuning and weaken a generalisation claim [30].

An independent test should sample new mechanisms across electronics and control, mechanical and thermal systems, distributed software, and reliability or manufacturing. Task authors should check the formal and engineering answers independently, including underdetermined cases. Match model class and tool access for EAL and the equal checker; score raw source, instruction-only, model-requested assessment, host-owned status and independent checker routes. Freeze task genealogy, model snapshots, prompts, budgets, status and explanation rubrics, source-review criteria and an information target before revealing outcomes. Measure candidate selection separately from exact family resolution. Retain failed calls, source revisions and human authoring/review effort. Such a study could support a model- and task-bounded causal delivery claim if assignment, measurement and coverage checks hold. A notation-specific claim additionally requires independently authored equal representations and comparable operations. External engineering cases would still be needed for field-use claims.

7. Conclusion

EAL/2 supplies a finite, executable account of named argument dependencies, typed method results and targeted objections. Its retained model trials show that already computed conclusions can be communicated more accurately than raw records on selected cases, while raw evidence and irrelevant-record handling remain unreliable for the tested recipient. Strict EAL raw-only correctness was 4/10 against 5/10 for meaning-equivalent JSON. The newer host owns the final claim status, limits a recipient to granted claims and resolves only reviewed family bindings. Two developmental cross-model cohorts measured recipient status consistency, malformed answers, model tokens, API latency and configured-rate charges; their checked final statuses were oracle-matched by construction. The initial request arm had a prompt conflict; a corrected route delivered 51/63 consistent statuses but only 11/63 strict agent-judged faithful explanations. The synthetic family-selection calls exposed two task-level false supports despite exact grants. An exploratory agent review flags explanation incompleteness, while human-

validated fidelity, field retrieval accuracy and amortised total cost remain unmeasured. An independent task sample and balanced equal-checker comparison are required to establish a language-specific or general reasoning advantage.

Data and code availability

The historical relay and notation results use repository revision 2b0815f; the later host, family, checker and offline schedules began at revision d78836b of <https://github.com/emmett08/earl>. The two 48-call pilot archives are under `benchmarks/results/2026-09-24-artifact-live-pilot/` and `benchmarks/results/2026-09-24-artifact-live-replication/`. The initial offline freeze and stopped attempt are under `benchmarks/results/2026-09-24-cross-model-offline/`; separate revised live archives, their read-only audits and a stricter recipient-status-consistency analysis are under `benchmarks/results/2026-09-24-cross-model-live/` (`stricter-read-only-analysis.tar.gz`). Their exact input SHA-256 values are 3d6a33d88720772f63255a1fac1210c05717b08fe4b49d7900a176920939529d and b8f9cf8fa03ac03c446c94d4e32287edbc0f45fcf4360c00195046375b8e13ef. The corrected route’s terminal ledgers and independent audits are under `benchmarks/results/2026-09-24-cross-model-route-v2/`; its complete 63-output exploratory review is under `benchmarks/results/2026-09-24-cross-model-route-v2-explanation-review/`. The separate de-identified 105-output cross-arm packet and AI reviews are under `benchmarks/results/2026-09-24-cross-model-explanation-review/`. The synthetic Kubernetes argument, application gate and offline family study are under `examples/kubernetes-resource-revision/` and `benchmarks/experiments/kubernetes-host-revisions/`. The distinct paid candidate-selection archive and its audit are under `benchmarks/results/2026-09-24-kubernetes-family-selection/`, frozen at SHA-256 b5ef86e16f16d539d75d0632497b73862a6f6ed4c1ea9cd42d6fdb313244c7cc. The archived notation analysis can be rerun without model calls using `scripts/analyse_notation_transfer.py` and `scripts/analyse_notation_format_sensitivity.py`. New live records and Kubernetes revisions are in PR 7 after d78836b; a final merge revision and public archive remain to be assigned. The repository was private when this manuscript was prepared; external reviewer access remains to be arranged by the author.

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